INCH-POUND

MIL-PRF-7032/2(AS)
20 June 2019

PERFORMANCE SPECIFICATION SHEET

INVERTER, AIRCRAFT, DC TO AC, 600 VA 1-PHASE, 400 HERTZ, MULTI-PURPOSE

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-7032.

SCOPE

1. Inverter. The inverter shall provide the electrical performance over a temperature altitude range of -55 °C to +71 °C and to 20,000 ft. altitude and as modified herein.

2. Part or identifying number (PIN). Inverters covered by this specification sheet are identified using the following PIN:

Static inverter – M7032/2-1 (see figure 1).

REQUIREMENTS

1. Description. Power ratings and characteristics of the inverter covered by this specification sheet are shown in table I.

TABLE I. Inverter description.

Part number	Power rating 1/	Power factor	Max weight (lbs)	Efficiency
M7032/2-1	600VA, 1Φ	0.75 MIN LAG 0.85 MIN LEAD	9	75% minimum

1/ Φ = Phase

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1. MIL-PRF-7032 variance: The inverter shall comply with MIL-PRF-7032, except that the following paragraphs of MIL-PRF-7032 shall be modified.

3.4.2.1.1 External electric connections. Delete the last sentence and replace with the following: "Input and output connections for single-phase operation shall be in accordance with table II of this specification sheet."

TABLE II. Connector pin assignments.

CONNECTOR	PIN	DESCRIPTION
J1	A	Spare
	B	28 VDC Return
	C	Spare
	D	+28 VDC
	E	Spare
	F	Spare
J2	A	Phase A Output
	B	Spare
	C	Spare
	D	Output Return
	E	Chassis Ground
	F	Chassis Ground
	G	Inverter Thermal Shutdown Indicator
	H	Inverter On Indicator
	J	Signal Ground
	K	Remote Switch Control
	L	Phase A Output
	M	Line Contactor Control
	N	Spare
	P	Spare
	R	Output Return
	S	Inverter On Indicator Return
	T	Inverter Thermal Shutdown Indicator Return
	U	Spare

3.4.2.1.1.2 Connectors. Delete in its entirety and replace with the following: "The part number of connector J1 shall mate with D38999/26WE6SN as specified in MIL-DTL-38999/26 and oriented as per figure 1. The part number of connector J2 shall mate with D38999/26WD18PN and oriented as per figure 1."

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3.4.2.1.1.3 Output voltage jacks. Delete in its entirety.

3.4.3.5 Cooling. Delete the fourth and fifth sentence and replace with the following: "The use of an internally mounted, integral cooling fan is permitted (see 6.6.2.5). The direction of the airflow is as shown on figure 1 of this specification sheet."

3.4.3.8 Size and weight. Delete the first sentence and replace with the following: "The size and dimensions of the inverter, including tolerances shall be as shown on figure 1 of this specification sheet. Inverter weight shall be as required in table I of this specification sheet."

3.5.1.1.a Delete and replace with: "The normal input voltage range is 22 to 29 volts DC."

3.5.1.1.b Delete and replace with: "The abnormal input voltage range is 20 to 31.5 volts DC."

3.5.1.1.c Delete and replace with: "The low input voltage range is 16 to 29 volts DC."

3.5.1.4 Efficiency. Delete in its entirety and replace with the following: "The minimum efficiency of the inverter shall be 75 percent."

3.5.1.7.1 Static inverters. Delete the first sentence and replace with the following: "The output voltage during input power application shall be within and remain within the limits of 3.5.1.2 of this specification sheet within two seconds after application of input voltage.

3.5.1.8.1 Overload. Delete in its entirety and replace with the following: "Over the normal input voltage range (see 3.5.1.1a), the inverter shall, without damage, maintain the minimum output voltage specified by 3.5.1.2a (1) while delivering 1000VA for 5 seconds and 750VA for 5 minutes."

3.5.1.8.2 Short circuit. Delete in its entirety and replace with the following: "Over the normal input voltage range (see 3.5.1.1a), the inverter shall be capable of delivering, without damage, at least 1200VA for 5 seconds."

3.5.1.8.3 Protection (static inverters only). Delete in its entirety and replace with the following: "The inverter shall have automatic electronic short circuit protection when the load is higher than the short circuit condition as specified in 3.5.1.8.2. The output voltage and/or current may drop to any level after the initial five seconds without damage to the inverter. The normal output voltage shall be automatically restored after the removal of the short circuit."

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Add the following new paragraphs to this specification sheet:

3.5.1.8.4 Inverter thermal shutdown indicator (Pin G of J2). An over-temperature status output signal of 28 volts DC and a maximum of 3.6 ampere shall be provided when the inverter temperature exceeds the operating requirements as specified in 3.5.2.1. The over-temperature status signal of 0 volts DC shall be provided when the inverter temperature resumes normal operating limits and the inverter shall resume normal operations. The output indication shall be in reference to the return indication at pin T of J2.

3.5.1.8.5 Inverter on indicator (Pin H of J2). An inverter ON status output signal of 28 volts DC and a minimum of 0.5 to maximum of 3.6 amperes shall be provided once the conditions of 3.5.1.8.4 have been met. An inverter OFF status output signal of 0 volts DC shall be provided once the conditions of 3.5.1.8.4 have been exceeded, or the AC output is off. The output indication shall be in reference to the return indication at pin S of J2.

3.5.1.8.6 Remote Switch Control (Pin K of J2). The inverter shall enable the AC output when provided a path to the signal ground at pin J of J2 via the remote ON control switch. This input shall be capable of carrying up to 3.6 amperes.

3.5.1.8.7 Line contactor control (Pin M of J2). The line contactor control shall provide 28 volts DC nominal and have a maximum current carrying capacity of 3.6 amperes. Short circuit automatic protection shall be provided for control output and normal operations will resume upon removal of short circuit. The control output shall turn on when the inverter output voltage is 112.5 – 117 volts RMS and 393 to 407 Hz as specified in 3.5.1.2a (1) and 3.5.1.3. The control output shall turn off when the required minimum voltage drops for 6 ±1 seconds or exceeds required voltage for 20 ±1 milliseconds. The control output shall turn off when the required frequency exceeds required levels for 1 to 3 seconds. The output control shall be in reference to the return control at pin J of J2.

3.5.1.8.8 Chassis ground (Pins E and F of J2). The DC resistance from the connector pins E and F to the inverter's chassis shall not exceed 50 milliohms.

3.5.1.8.9 Current returns. The AC neutral and DC current return terminals shall be connected to its own current return wire. The current return wiring shall be separate from any wires used to bond internal components to the chassis of the inverter.

3.5.1.8.10 Electrical bonding. The bonding resistance between the mounting surface and any part of the inverter shall comply with MIL-STD-464, section 5.11. The bonding resistance between the mounting surface and each connector case shall be not greater than 2.5 milliohms.

3.5.1.8.11 Safety. System safety design and operational safety requirements shall be in accordance with those applicable requisites presented in MIL-STD-882, Section 4.3 and MIL-STD-1472, Section 5.13.

3.5.1.9 Electromagnetic compatibility. Delete the first sentence and replace with the following: "The inverter shall conform to the requirements of MIL-STD-461 and MIL-STD-464, when tested to the requirements as specified in 4.5.8."

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3.5.1.16 Acoustical noise. Delete in its entirety.

3.5.2.1 Temperature altitude. Delete the paragraph in its entirety and replace with the following: "The inverter covered by this specification sheet shall be capable of operation without degradation of performance throughout the temperature altitude range of -55 °C to +71 °C and to 20,000 ft. altitude (see 4.5.2.3 and 4.5.2.4)."

3.5.2.8 Dust. Delete in its entirety and replace with the following: "The inverter shall withstand the effects of sand and dust from world-wide operation, as simulated by the dry dust test of 4.5.23."

3.7.1 Terminals. Delete the paragraph in its entirety and replace with the following: "The input and output connections and pin assignments shall be marked as shown on table II and figure 1 of this specification sheet (see 4.5.1)."

3.7.3 Nameplate. Delete the paragraph in its entirety and replace with the following: "A nameplate conforming to MIL-DTL-15024, type A, B, C, G, or H, with the following markings, in addition to the requirements for MIL-STD-130, shall be securely attached to the inverter. The nameplate shall contain the following information (see 4.5.1):

Inverter 600 VA
Input: ___ Volts DC ___ Amperes DC
Output: ___ Volts AC ____Amperes AC
400 Hz Single Phase
Military Part No. M7032/2-1
National Stock No.
Manufacturers Part No.
Manufacturers Serial No.
Date of Manufacture
Contract or Order No.
Manufacturer's name or trademark
Inspector's stamp
US

3.7.4 Airflow. Delete first sentence and replace with the following: "When an integral cooling fan is used, direction of airflow shall be plainly and permanently marked on the outside of the housing in the direction specified on figure 1 of this specification sheet."

3.7.6 Output voltage jacks. Delete in its entirety.

VERIFICATION:

4.2.2 Qualification inspections and tests. Add the following new second sentence: "Testing shall be in accordance with MIL-PRF-7032, as modified by this specification sheet."

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4.2.3 Qualification retention. Add the following new second sentence: "Testing shall be in accordance with MIL-PRF-7032, as modified by this specification sheet."

4.4.1.6 Warm-up and stabilization. Delete the first sentence in its entirety and replace with the following: "Prior to testing, the inverter shall be operated, delivering full rated load as required in table I of this specification sheet for sufficient time for the inverter's temperature, output voltage, and frequency to stabilize (see 6.5.5)."

4.5.1 Visual and mechanical inspection. Add the following to the end of the paragraph: "Inverter weight shall be as required in table I of this specification sheet. Inverter dimensions and markings shall be as shown on figure 1 of this specification sheet."

4.5.2.2 Power factors. Delete the first sentence and replace with the following: "The tests specified in 4.5.2.1 shall be repeated at the minimum lagging and minimum leading power factors in table I of this specification sheet."

4.5.8 Electromagnetic compatibility. Delete in its entirety and replace with the following: "The inverter shall be tested and be in accordance with the requirements and applicable limits of MIL-STD-461. All inverters shall meet the requirements of CE102, CS101, CS114, CS115, CS116, CS117, CS118, RE102, and RS103. All output leads, including the neutral, shall comply with the requirements of CE102. Inverters used in Army or Anti-submarine Warfare (ASW) applications shall also meet the requirements of CE101, RE101, and RS101. The test shall be conducted at full load and half load, at minimum and maximum normal input voltage. The inverter load power factor shall be adjusted such that the combined power factor of the load and all auxiliary test equipment required shall present a total load of unity power factor to the inverter. During electromagnetic susceptibility testing, the output voltage and frequency shall remain within the normal limits of 3.5.1.2 and 3.5.1.3. Additionally, the inverter shall comply with the bonding requirements of MIL-STD-464, Section 5.11."

Add the following new subparagraph 4.5.8.1:

4.5.8.1 External electromagnetic environment. As installed in the aircraft, the output voltage and frequency shall remain within the normal limits of 3.5.1.2 and 3.5.1.3 while the inverter is exposed to a worst case composite of the MIL-STD-464 tables I, II, and V electromagnetic environments.

4.5.10.1.1.b Linear loads (full wave sinusoidal output current). Add the following sentence after the first sentence: "Inverter full rated load values shall be as required in table I of this specification sheet."

4.5.15 Acoustical noise. Delete in its entirety.

4.5.17 Acceleration. Delete in its entirety and replace with the following: "The inverter shall be tested in accordance with the acceleration test of MIL-STD-810, Method 513, Procedure II. The required acceleration level for all directions in procedure II (operational test) shall be 6.0g, and shall be applied along three mutually perpendicular axes in two opposite directions along each axis. The test time duration in each direction shall be at least one minute following

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centrifuge stabilization, and the item shall be operated continuously at 28 VDC input and full load during the test. The output voltage and frequency shall be monitored during the test to determine compliance with 3.5.2.2. At the conclusion of the test, the inverter shall be examined for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c."

4.5.19 Shock. Delete in its entirety and replace with the following: "The inverter shall be subjected to functional shock (Procedure I) and crash hazard shock (Procedure V) tests in accordance with MIL-STD-810, Method 516.7 to determine compliance with 3.5.2.4. Calibration shocks shall be applied in accordance with MIL-STD-810, Method 516.7 prior to shock testing.

4.5.19.1 Functional. Delete in its entirety and replace with the following: "The inverter shall be operating during the functional shock test at full load and 28 VDC input. The output voltage shall be monitored during the test to determine compliance with 3.5.2.4.1. The applied shock response spectrum (SRS) shall conform to the Functional Test for Flight Equipment spectrum of Figures 516.6-8, 516.7-9, and Table 516.7-III of MIL-STD-810. The inverter shall be subjected to three shocks in each direction along three mutually perpendicular axes of the test item for a total of 18 shocks at the functional level. At the conclusion of the functional shock test, the inverter shall be inspected for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c."

4.5.19.2 Crash hazard. Delete in its entirety and replace with the following: "The inverter shall not be operated during the crash hazard shock test. The applied SRS shall conform to the Crash Hazard Test for Flight Equipment spectrum of Figures 516.6-8, 516.7-9, and Table 516.7-III of MIL-STD-810. The inverter shall be subjected to two shocks in each direction along three mutually perpendicular axes of the test item for a total of 12 shocks at the crash hazard level. At the conclusion of the crash hazard shock test, there shall be no failure of the mounting attachment, the test item shall remain in place, and no parts shall have separated from the inverter. However, bending and distortion of the inverter shall be permitted."

4.5.20 Vibration. Delete in its entirety and replace with the following: "The inverter shall be subjected to vibration testing of MIL-STD-810, Method 514.7, Procedure I (general vibration), Category 14 (rotary wing aircraft - helicopter). The inverter shall be operating during the vibration test at full load and 28 VDC input. The output voltage and frequency shall be monitored during the test to determine compliance with 3.5.2.5. The applied vibration reference profile shall be of MIL-STD-810, Method 514.7, Annex D, Figure 514.7D-3 and Table 514.7D-IIIb (general materiel) tailored to the AH-1 aircraft platform using the sinusoidal frequencies and magnitudes shown in table III of this specification sheet. Duration shall be 4 hours in each of three mutually perpendicular axes. Equivalence methods may be applied to test durations where applicable (i.e. amplitudes may be reduced and durations increased) per guidance provided in MIL-STD-810, Method 514.7, Annex F. The test article shall demonstrate specific performance during and upon completion of this test sequence and exhibit no structural damage or degradation. Pre- and post-test frequency response measurements shall be used to identify structural damage."

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TABLE III. SOR (Sine-on-Random) profile parameters.

Description	Hz	Amplitude
Main Rotor 1P (A_1 / f_1)	4.9	0.12 g
Main Rotor Blade Pass (A_2 / f_2)	19.7	1.97 g
2X Main Rotor Blade Pass (A_3 / f_3)	39.4	2.50 g
3X Main Rotor Blade Pass (A_4 / f_4)	118.1	1.50 g
W_0		0.001 g^2/Hz
W_1		0.010 g^2/Hz
f_t	500	

4.6.1 Temperature cycling. Delete the second sentence and replace with the following: "Twenty of the operating hours shall be at unity power factor, 15 hours at the minimum lagging power factor and the remaining 15 hours at the minimum leading power factor as required in table I of this specification sheet."

Referenced documents. In addition to MIL-PRF-7032, this specification sheet references:

ASME 14.5.1M	MIL-STD-461	MIL-STD-464
MIL-STD-810	MIL-STD-882	MIL-STD-1472
MIL-STD-130	MIL-DTL-15024	MIL-DTL-38999/26

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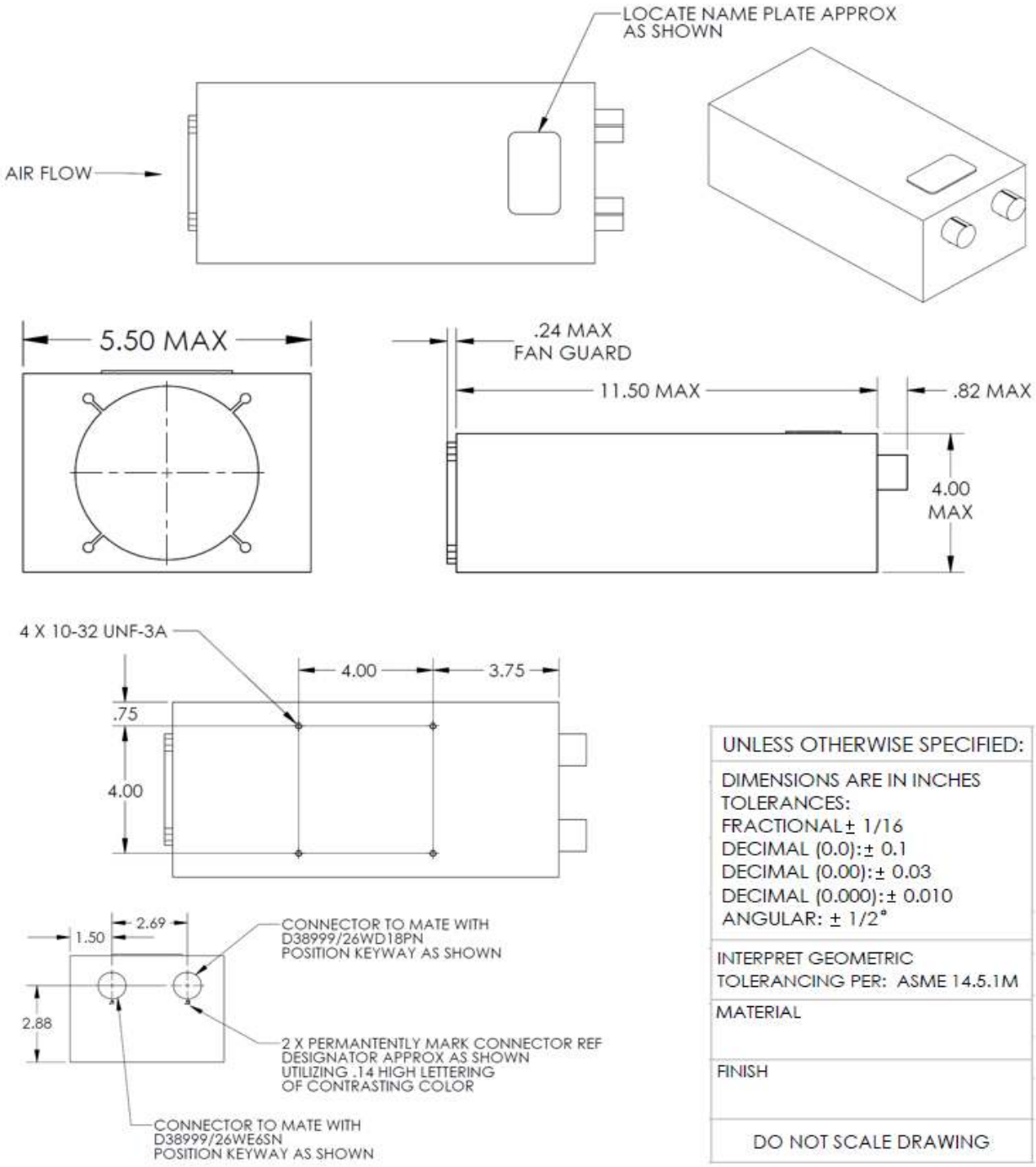


FIGURE 1. Static inverter M7032/2-1.

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CONCLUDING MATERIAL

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